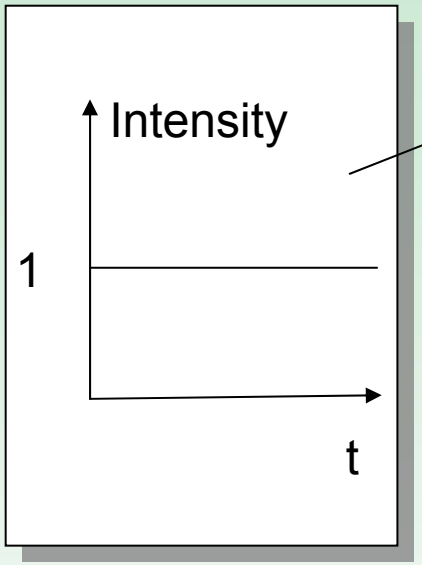
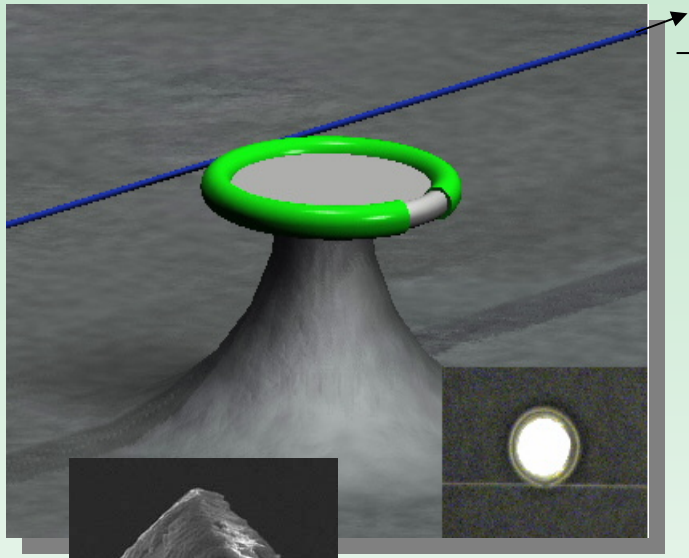
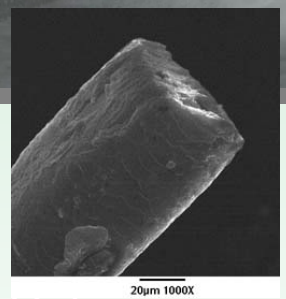
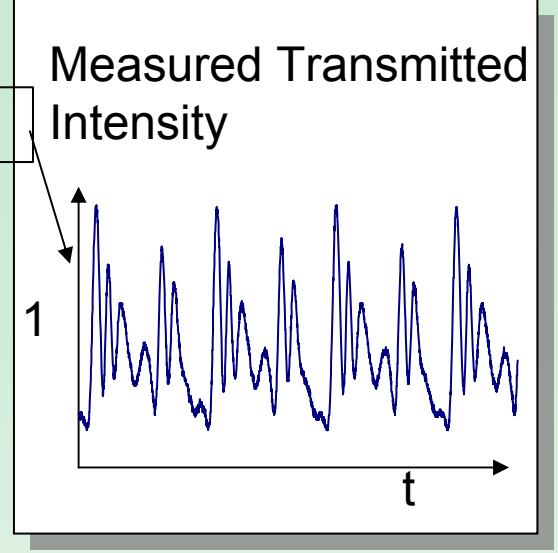
Radiation pressure induced vibrations in optical cavities









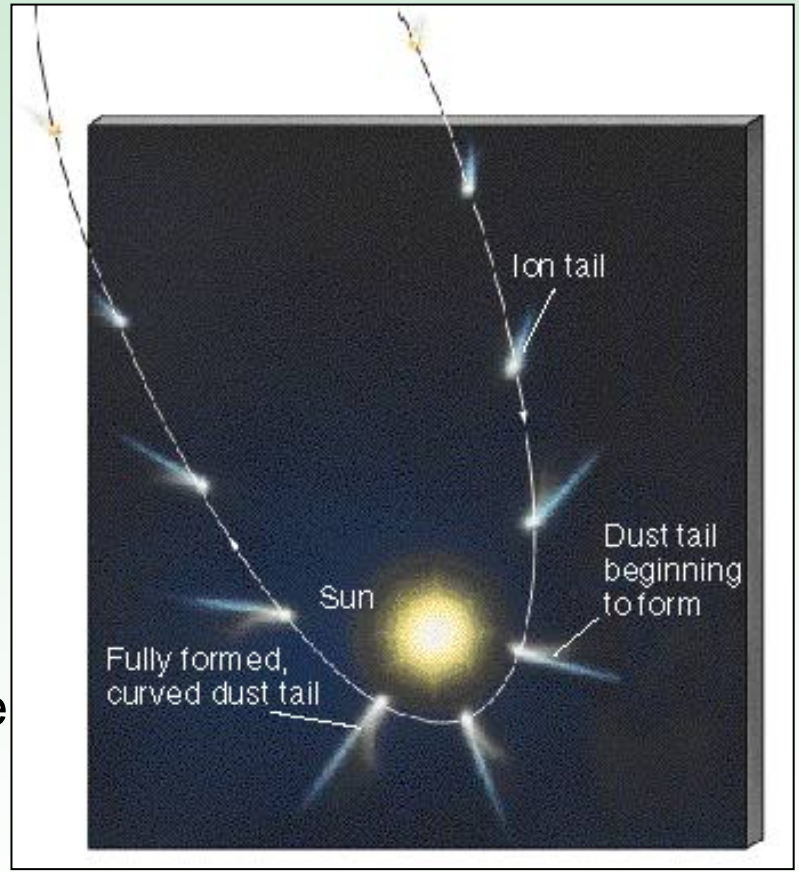
T. Carmon., H. Rokhsari, L. Yang, T. Kippenberg and K. Vahala
tal@caltech.edu
California Institute of Technology
Department of Applied Physics
http://www.vahala.caltech.edu/

Ein Bokek, Israel - February, 2005

FRISNO-8
FRench-Israeli Symposium on Non-linear & Quantum Optics

Who is interested in radiation pressure?

- First proposed by Kepler and later elaborated on by Newton.
- Bartoli infers the necessity of light pressure from thermodynamics.
- Maxwell



400 years ago, Kepler observed that comet's tail is opposing the sun, suggesting radiation pressure to be the reason.

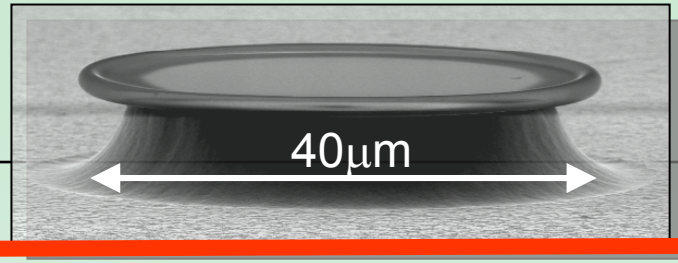
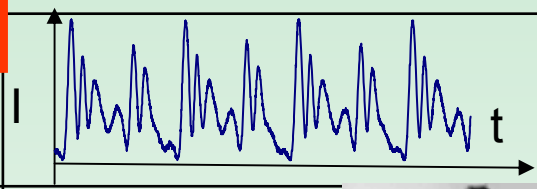
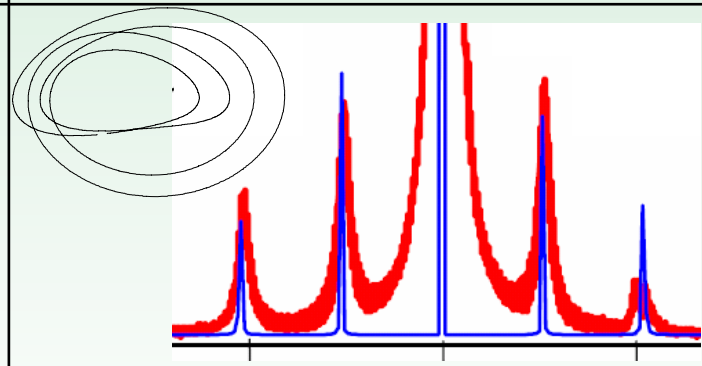
VOLUME 92, NUMBER 16 PHYSICAL REVIEW LETTERS week ending 23 APRIL 2004

Experimental Demonstration of a Classical Analog to Quantum Noise Cancellation for Use in Gravitational Wave Detection

C. M. Mow-Lowry, B. S. Sheard, M. B. Gray, and D. E. McClelland S. E. Whitcomb

- Laser Interferometer Gravitational-Wave Observatory
- Radiation pressure is the limit for gravitational wave measuring



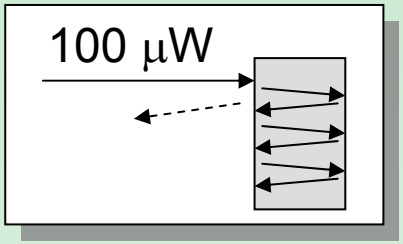
Why optical cavities of small volume and low losses are needed?	
How do we make them? Old New	
Experimentally, What effect do we see?	
Theory, Why (hand waving + model)	$\lambda\Delta\Phi$ 
Fine details of oscillations: a) From sine to chaos b) Alternating peaks number c) Fourier space	
Similarities with Kerr induced parametric oscillations.	Kerr, n (I) Radiation pressure l (I)
Size dependant, other cavities, what can we do with this (QM)?.	

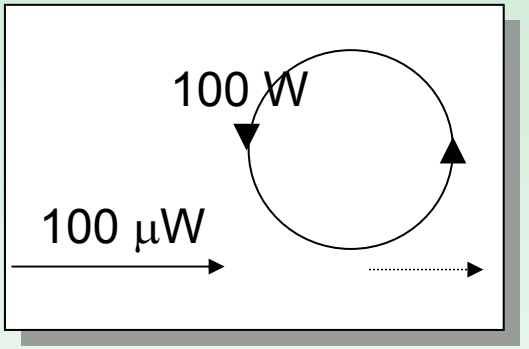


Why small volume and low optical losses?

- Have : 100 μW source,
- Need : 100 W/μm^2 .

We need more than we have

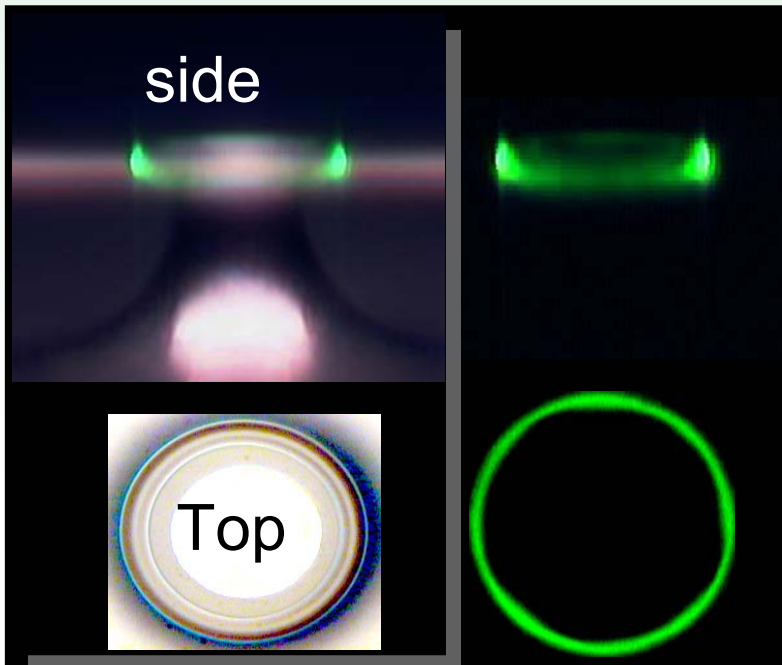




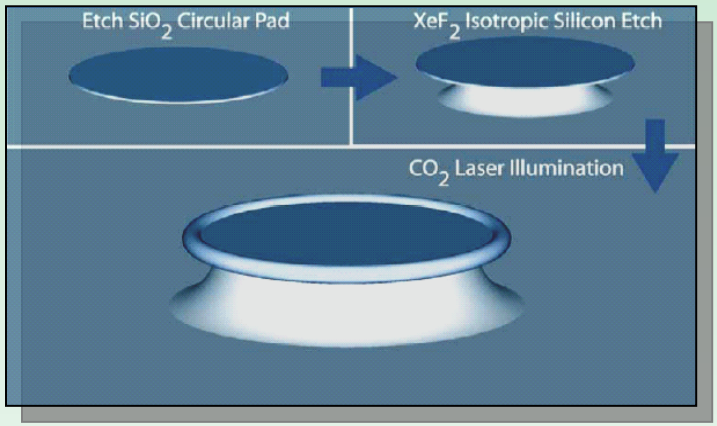
Low losses: each photon will pass more time via the same cross section

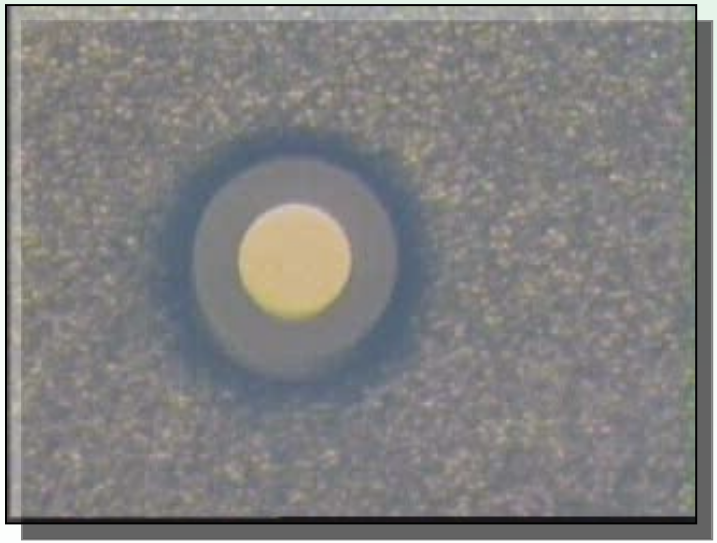
Small dimensions: small cross section

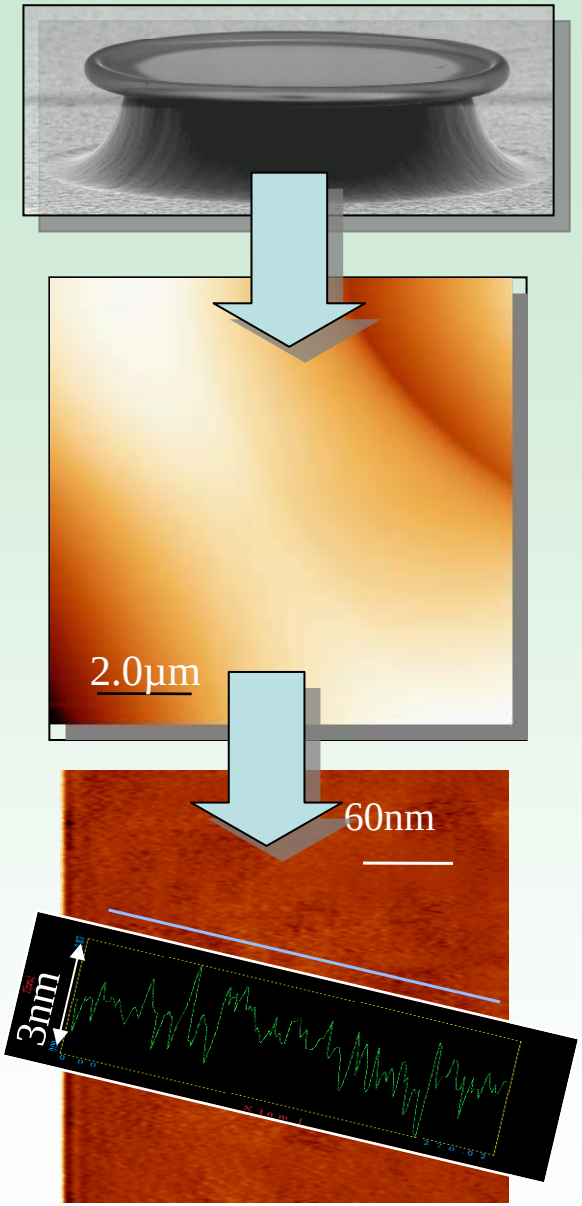
Low losses →
→ Narrow bandwidth of the optical resonance
→ Long photon lifetime (1 μs)
→ (100 meters)
→ (1,000,000 round trips)



Manufacturing







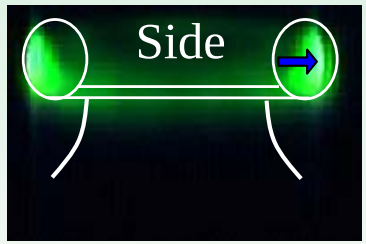
D. Armani, T. Kippenberg, S. Spillane and K. J. Vahala
"Ultra-high-Q toroid microcavity on a chip," Nature, 421, 925, 2003.

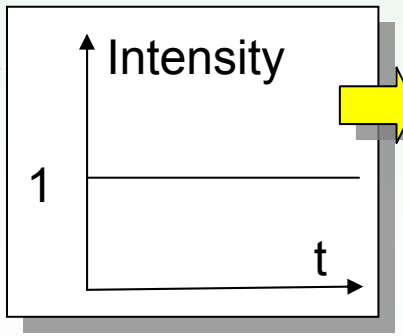
Miniaturization together with optical loss reduction in optical cavities improves key figures of merits for nonlinear optics

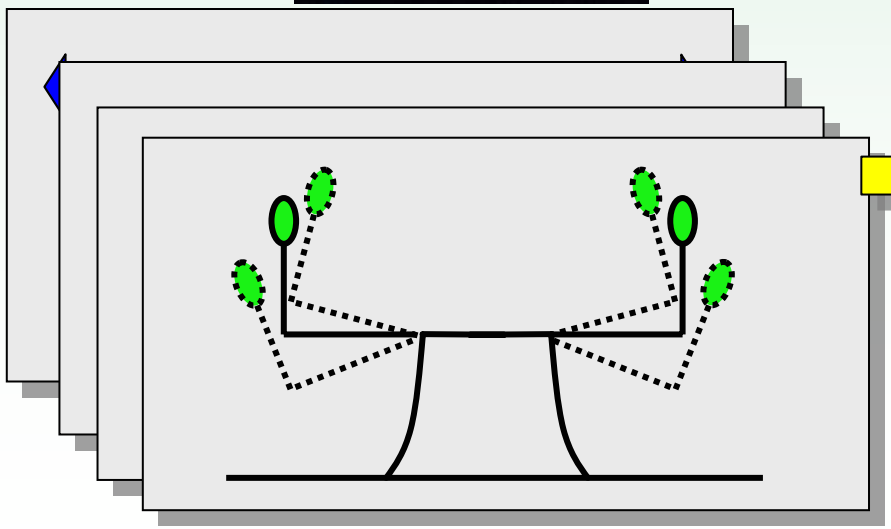
- Known

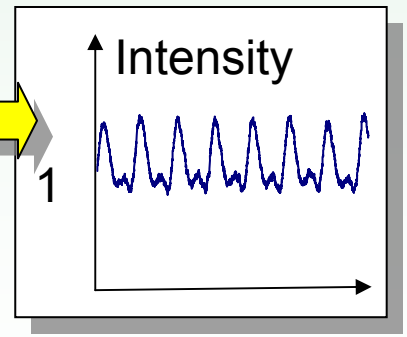
- What is new?

Miniaturization together with optical loss reduction → observation mechanical oscillations originates from the pressure of circulating radiation



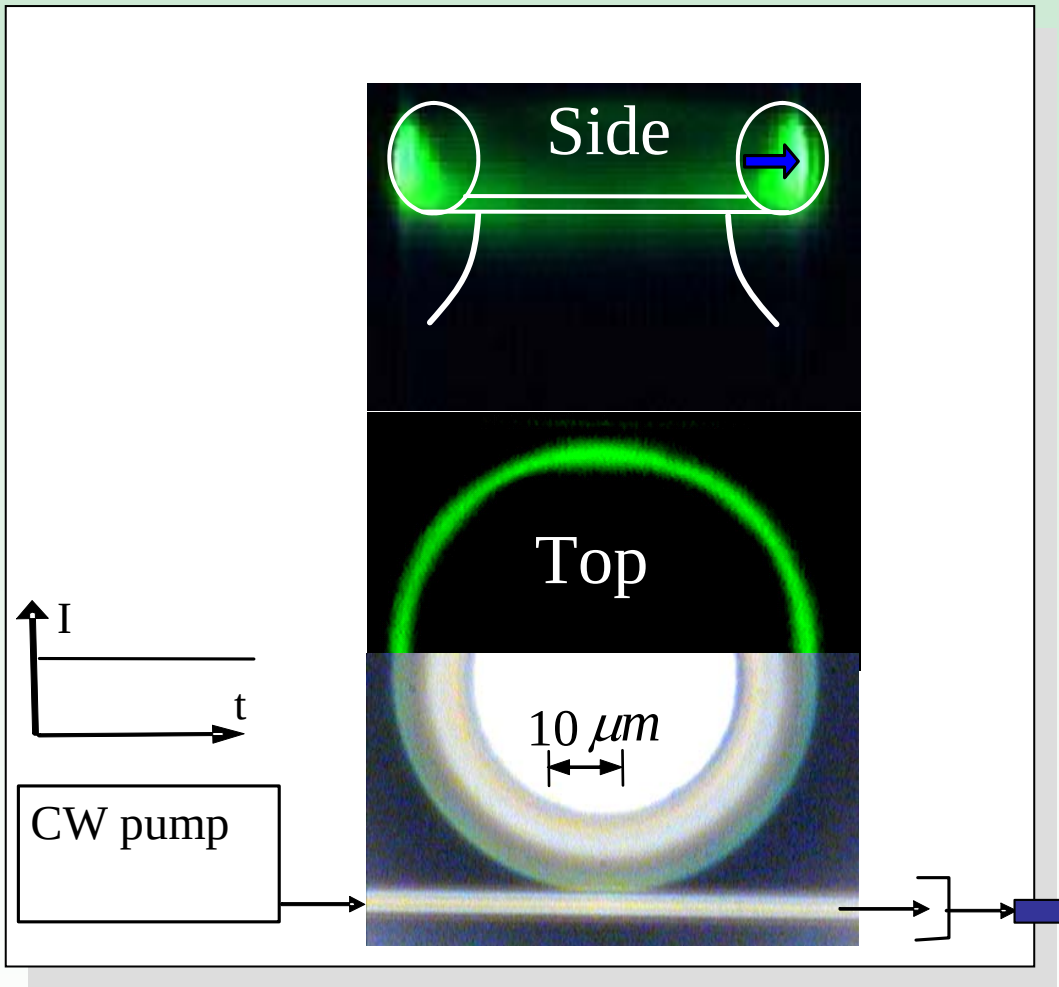




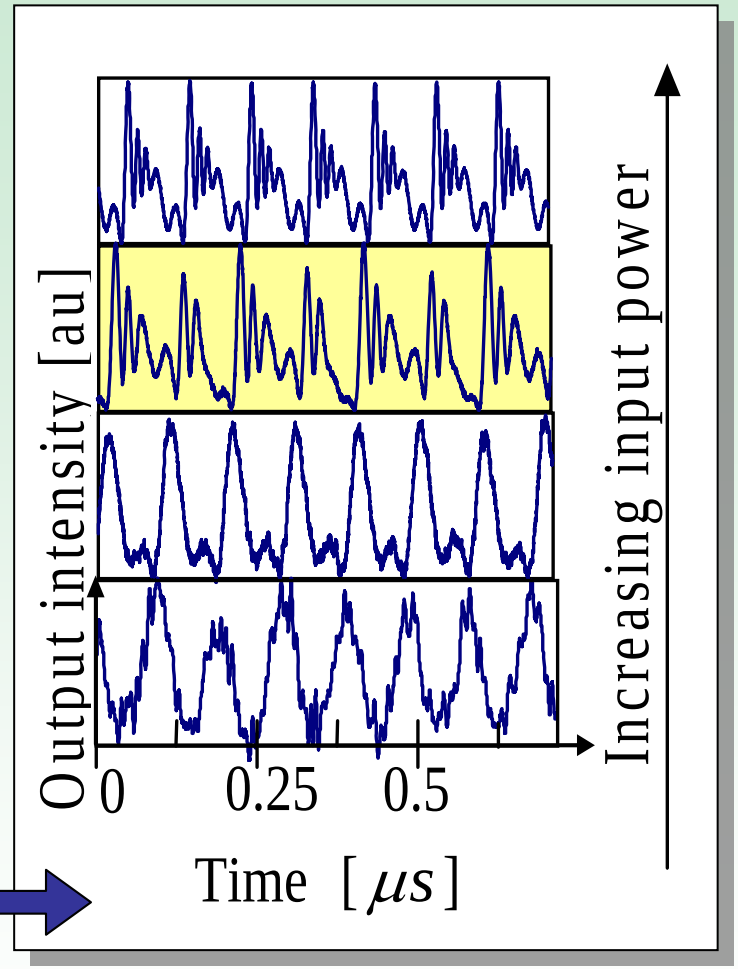


What experimental system do we have?, What do we see?

Experimental system

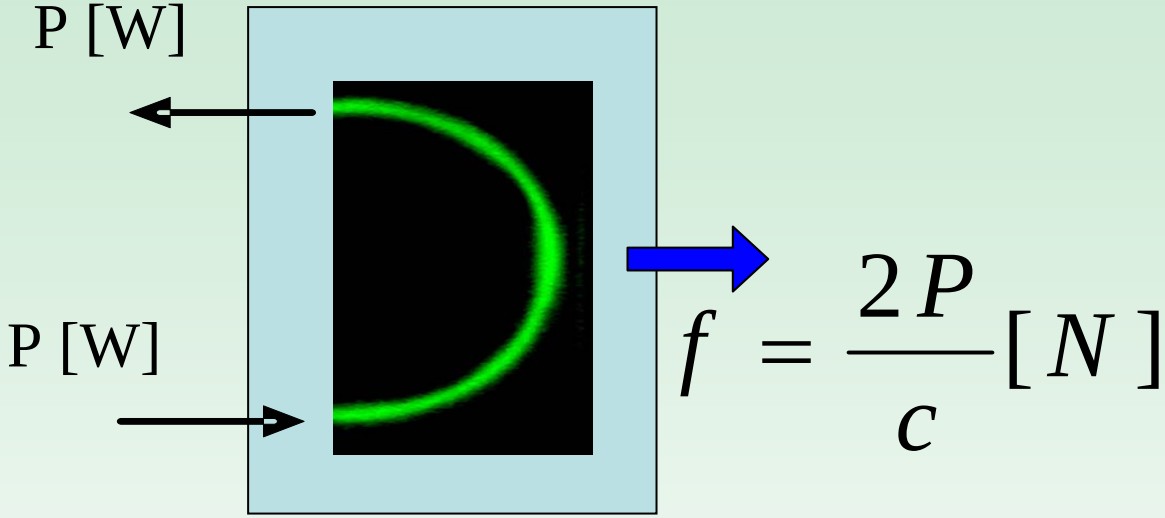


What do we see?



How does it look in the frequency domain?

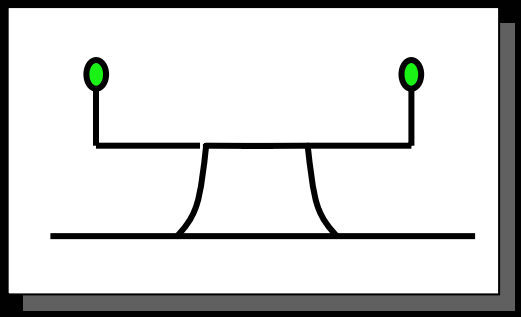
Why force?

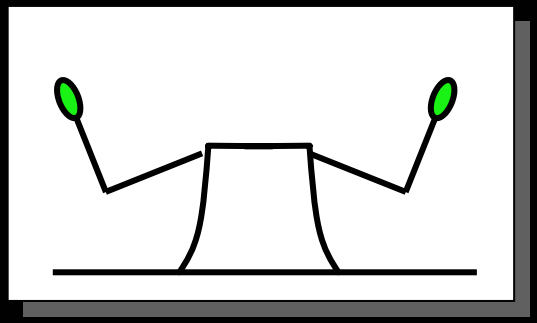


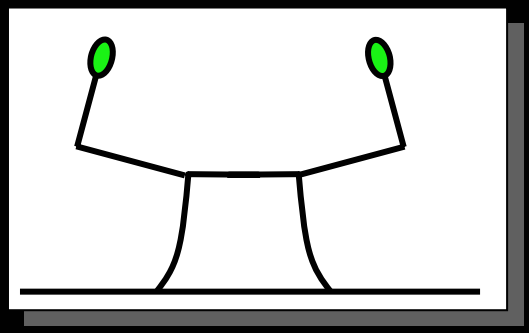
Fundamental principle of linear momentum conservation.

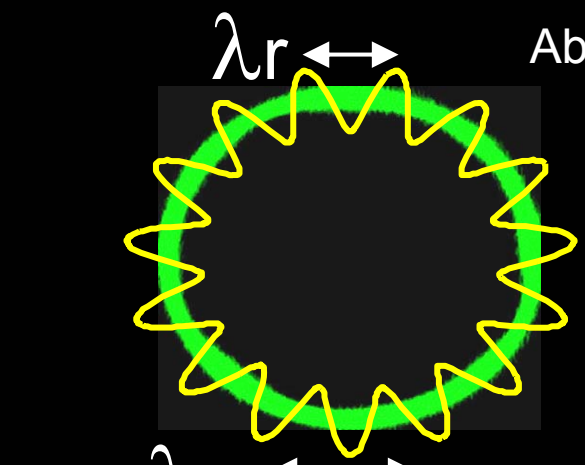
How to translate force to optical effect?

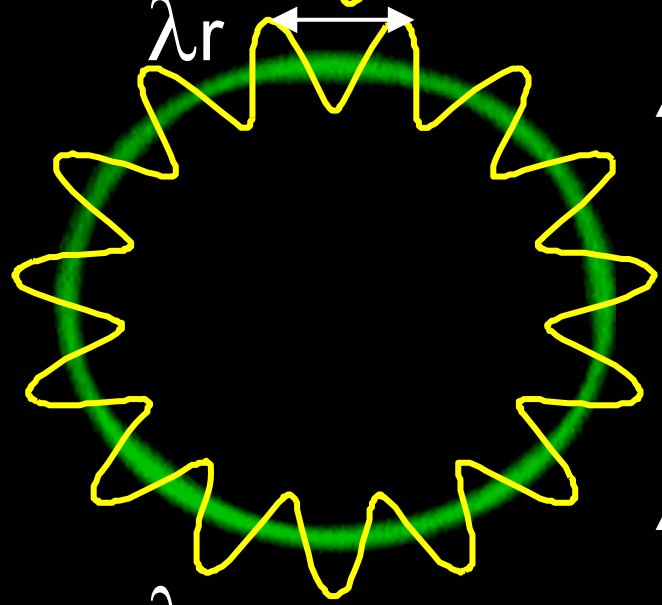
Circumference=n λ

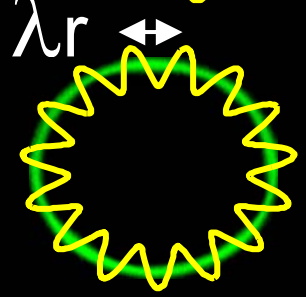


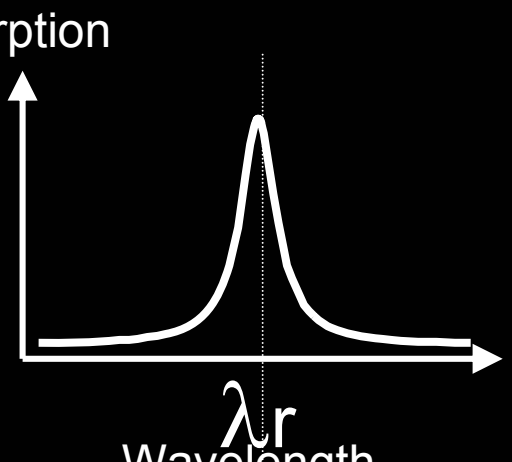


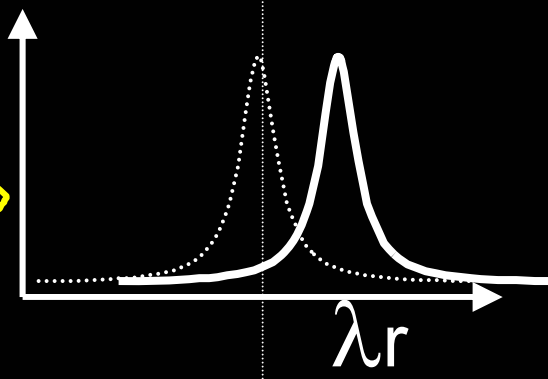


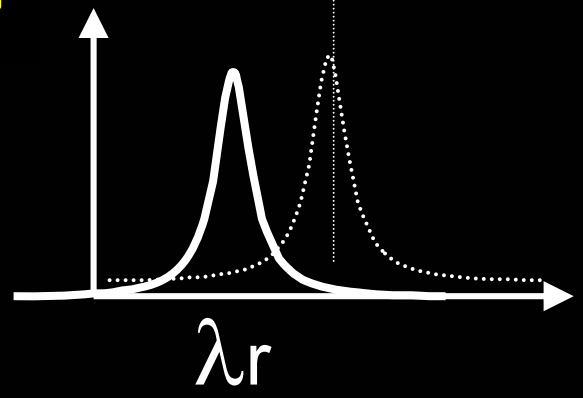




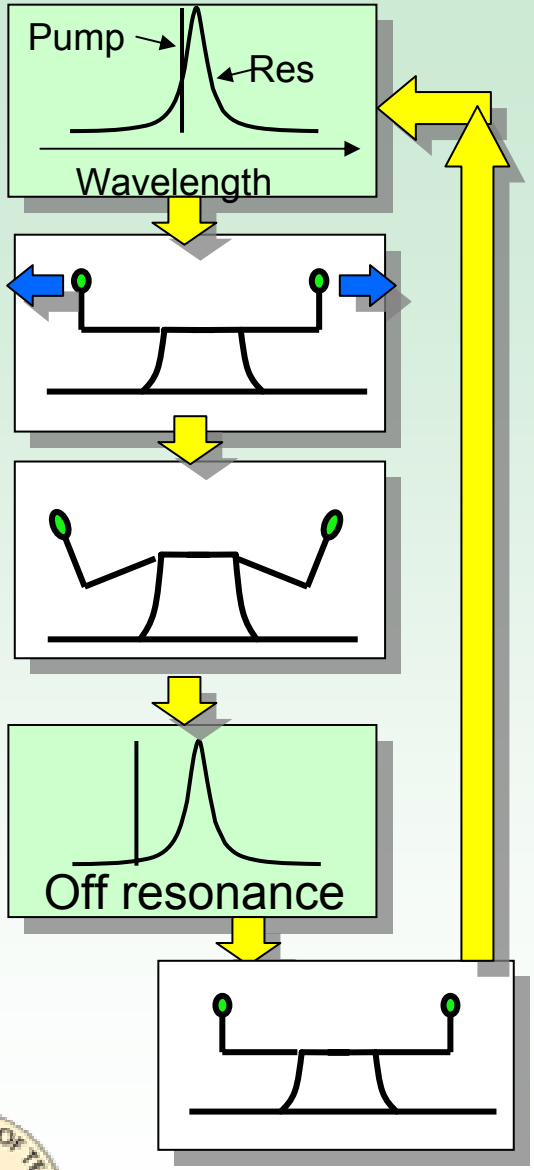


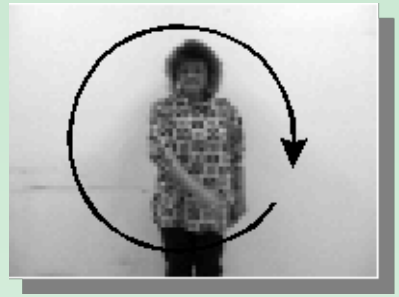






Pump wavelength<Resonance (It is important)





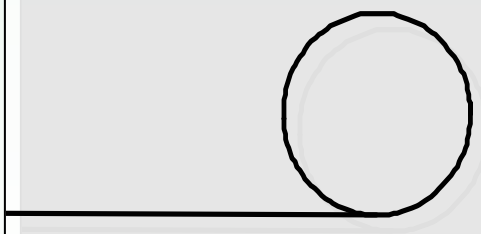
Energy transfer	
Inflation	Deflation
Optical->mechanical	Mechanical -> optical
Red Doppler shift	Blue Doppler shift

Regenerative oscillation

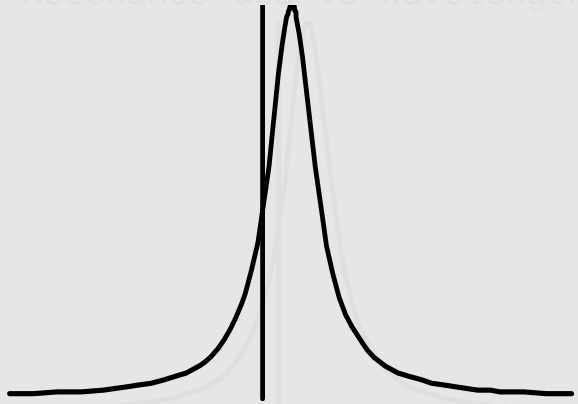
Low Amplitude vibrations



Structure's vibrations



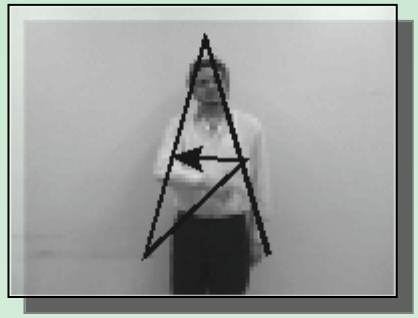
Resonance abs Vs Wavelength

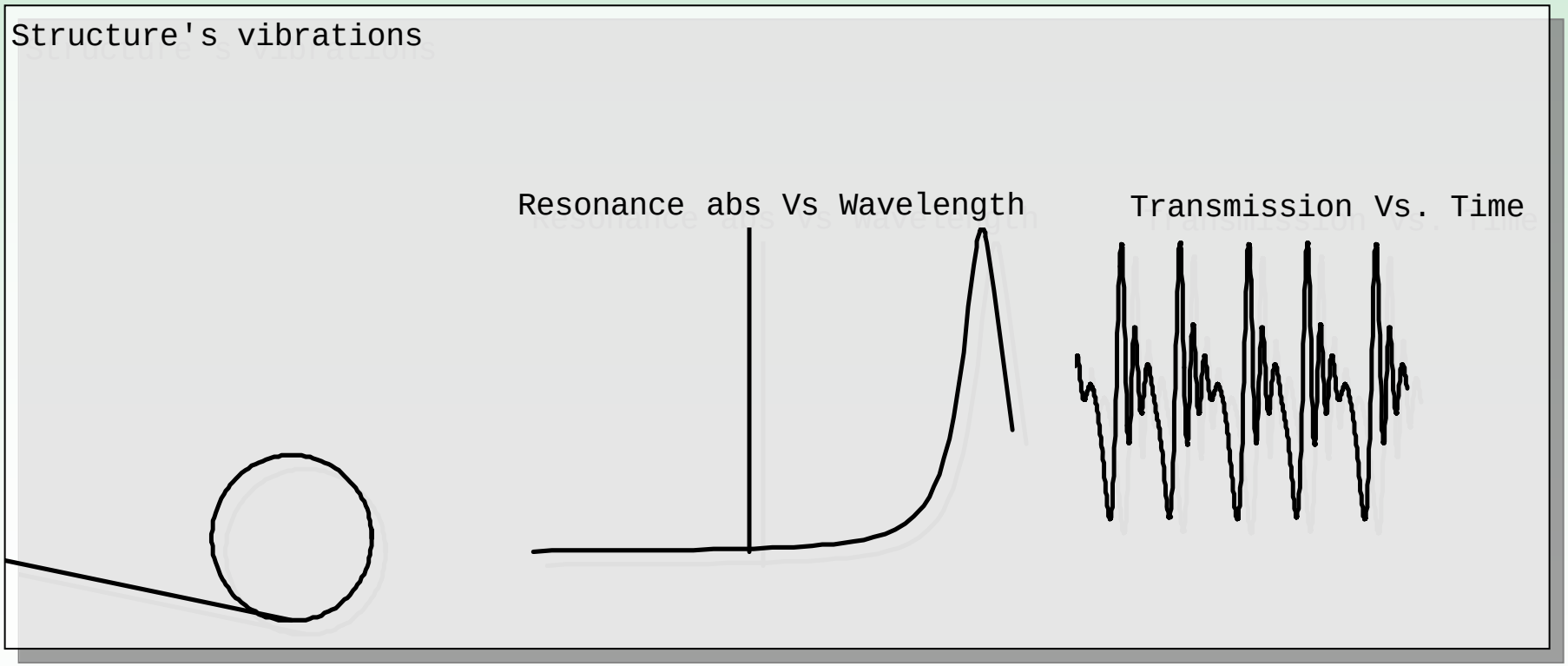


Transmission Vs. Time



Large amplitude vibrations

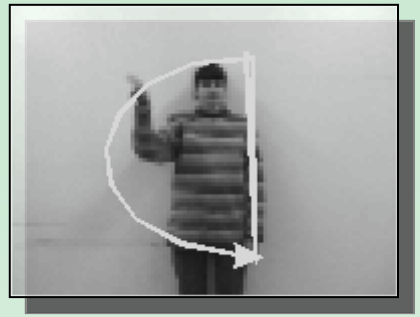




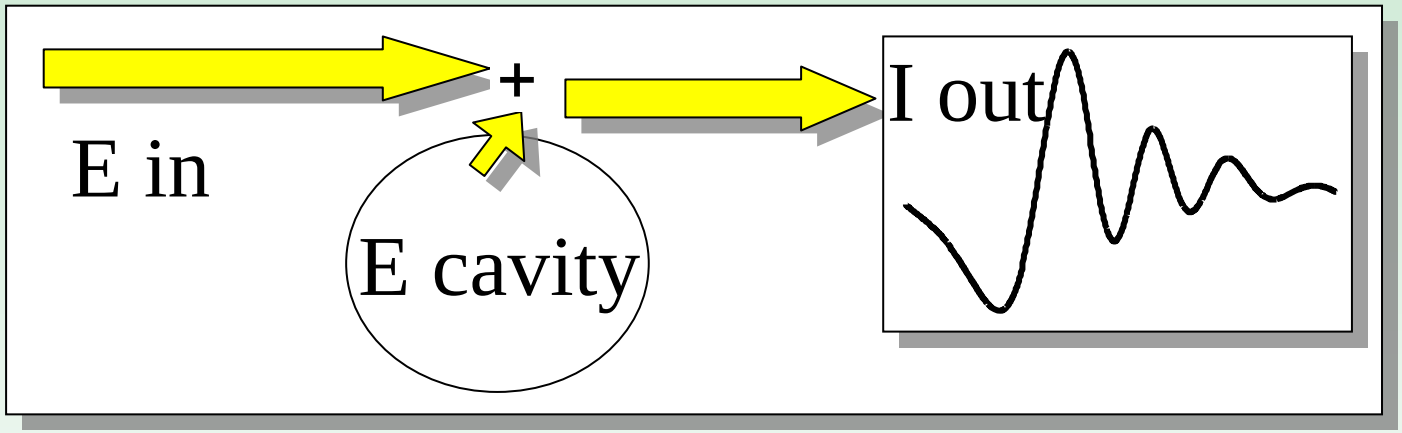
Why decaying peaks?

Lorentzian is only valid for steady state and not for a dynamical situation!

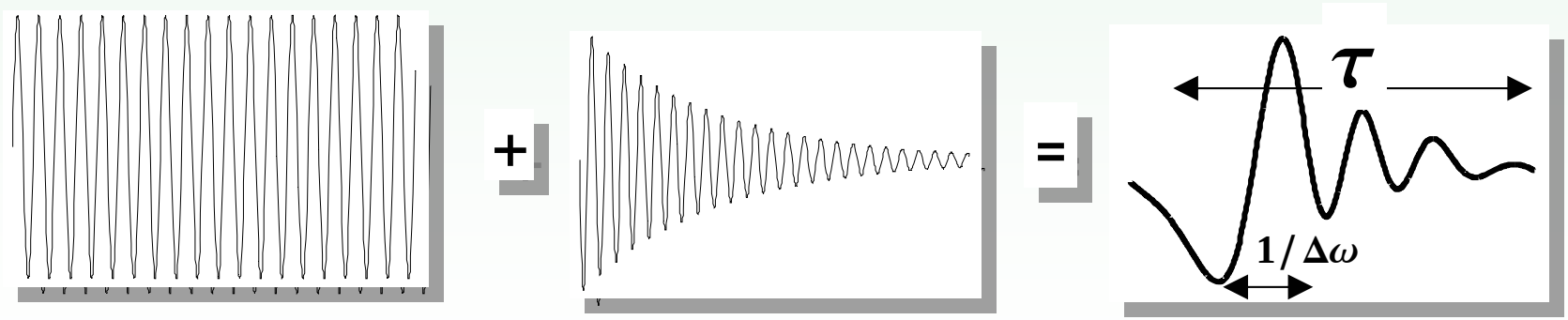
Why decaying peaks?



Lorentzian absorption is only steady state approximation

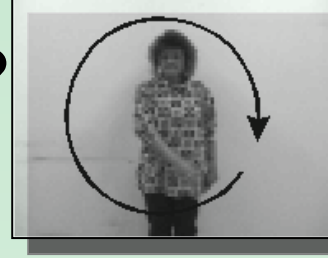


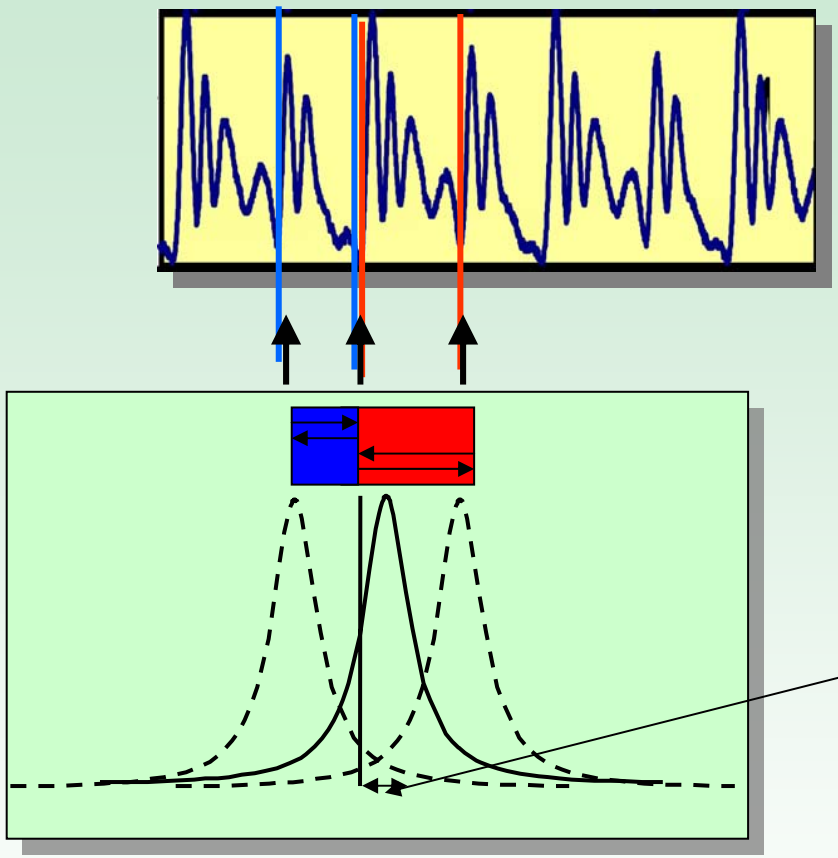
$$< \left(\sin(\omega t) + e^{-t/2\tau} \sin[(\omega + \Delta\omega)\ t] \right)^2 > =$$



Why alternating number of peaks (4, 3, 4, 3…)?

Why alternating number of peaks (4, 3, 4, 3…)?





•Cavity is charged with light upon passing of the absorption bell through pump wavelength

•And discharged later

•Resonance bell spends more time on one side of pump wavelength because of the initial offset (4-3=1 peak longer).

•Offset necessary for the red shift to be stronger than blue shift and energy to flow from optical mode to mechanical mode.

(negative offset = device cooling)

Cooling: •P. F. Cohadon, A. Heidmann, and M. Pinard, Phys. Rev. Lett. **83**, 3174 (1999).
•V. B. Braginsky, A. B. Manukin, and M. Y. Tikhonov, Sov. Phys. JETP **31**, 829 (1970).
•A. Dorsel. J.D. McCullen, P. Meystre, H. Walther, and E.M. Wright, Phil. Trans. R. Soc. Lond. A **313**, 341 (1984)

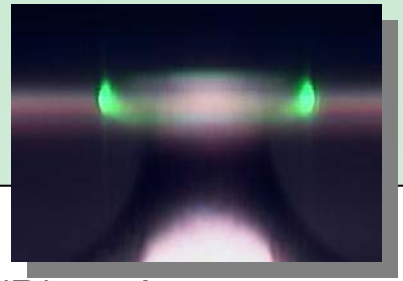
Stop waving with your hands,
we want dynamical equations





Model

no free parameters!



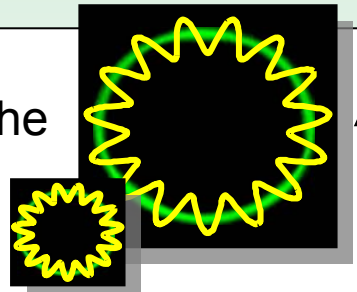


*Optical field inside cavity, A(t)
As function of pump amplitude (B) and pump resonance detuning ($\Delta\omega$)

$$\dot{A}(t) + A(t)[\alpha\, c/n - i\,\Delta\omega] = i\,B\sqrt{\alpha\, c/n\,\tau_0}$$

($\Delta\omega$: ?)

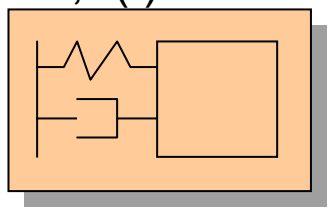
Pump-resonance detuning as function of the mechanical flex



$$\Delta\omega = \Delta\omega_0 - \omega\mu\, r(t)\, 2\pi\, n/(\lambda_0 N)$$

($r(t)$: ?)

Cavity deviation from equilibrium, r(t)
Forced harmonic oscillator.



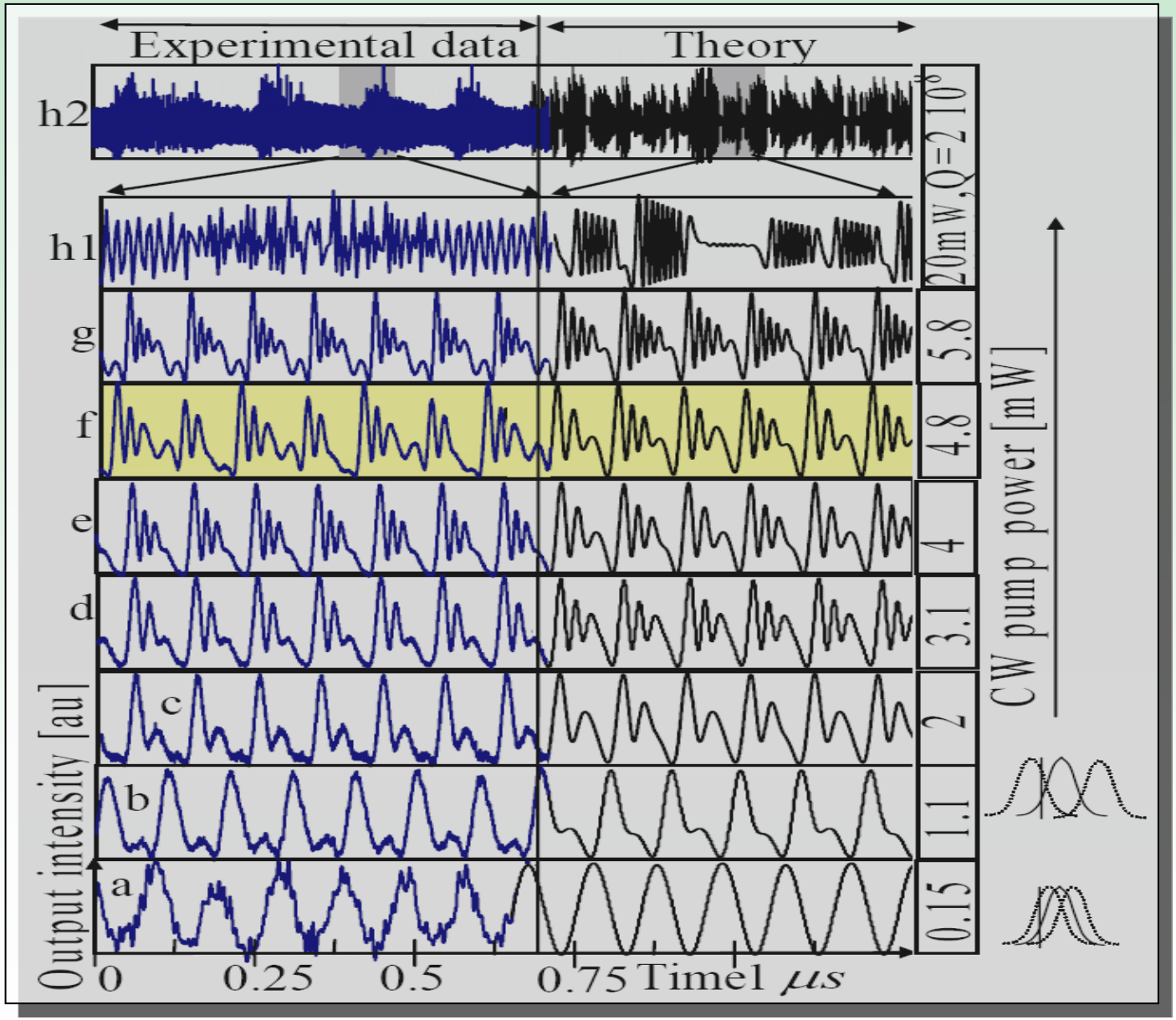
$$m\ddot{r}(t) + b\dot{r}(t) + kr(t) = \mu f(t)$$
$$= \mu 2\pi\, |A(t)|^2\, n/c$$

($A(t)$: ?)

Compare theory to experimental results

* M. L. Gorodetsky and V. S. Ilchenko, JOSA B, 16, 147 (1999)







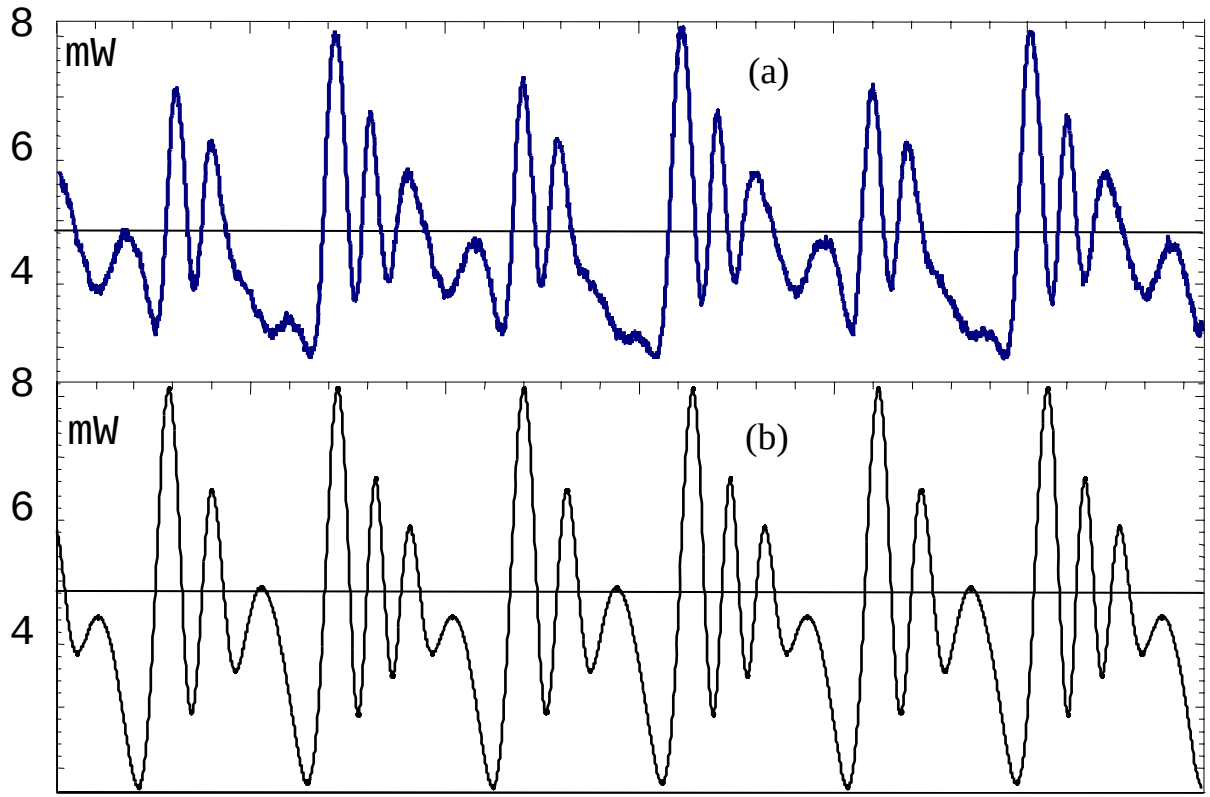
Tal Carmon, Hossein Rokhsari, Lan Yang, Tobias J. Kippenberg and Kerry J. Vahala. *Temporal behavior of radiation-pressure-induced vibrations of an optical micro-cavity phonon mode* , Accepted for publication in Phys. Rev. Lett.



Output Power

Measured

Calculated

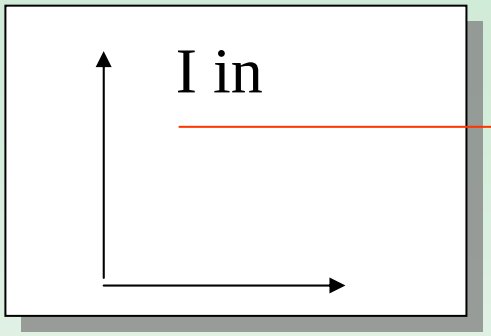


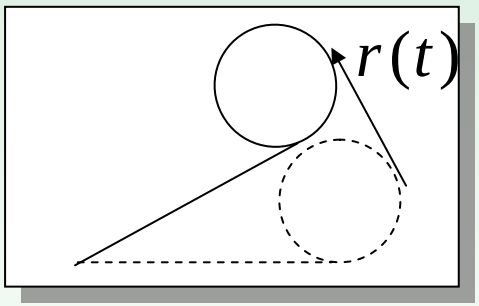


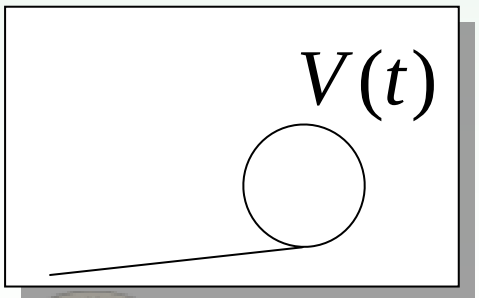
What is the mechanical dynamics? (position, velocity)

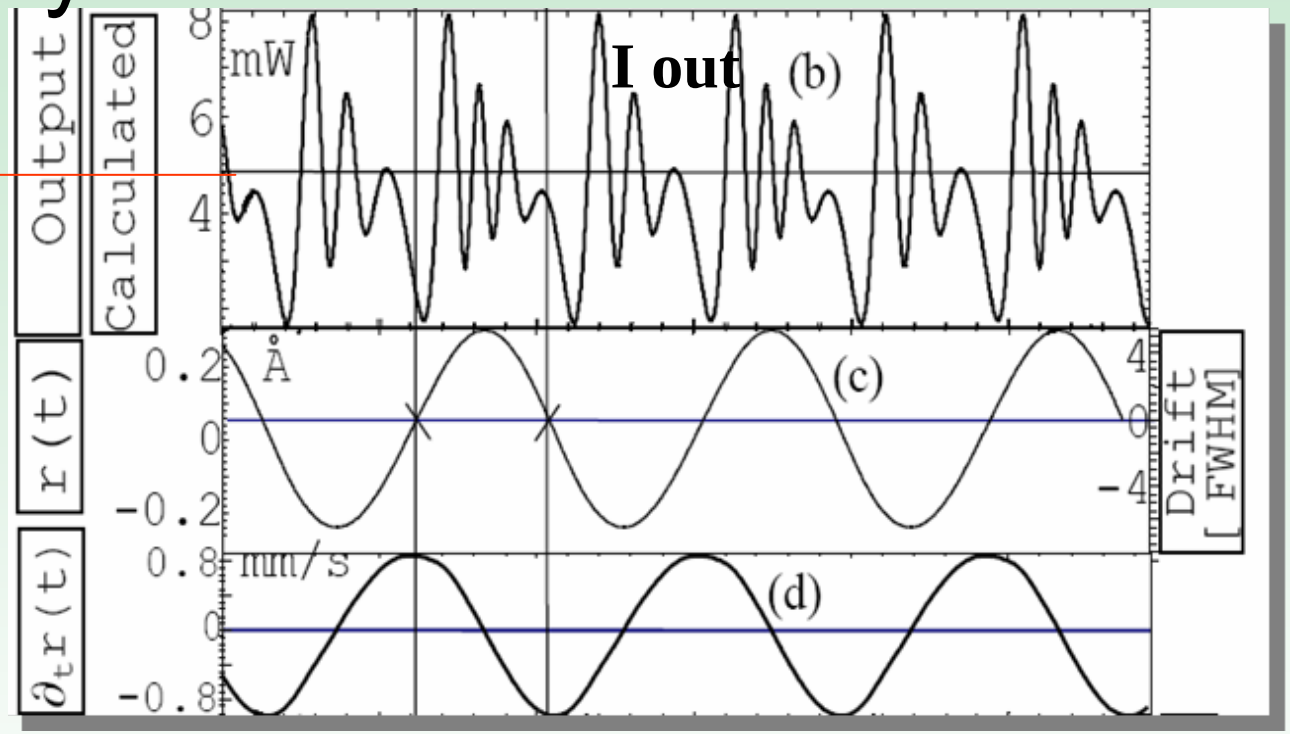


Mechanical dynamics, position, velocity









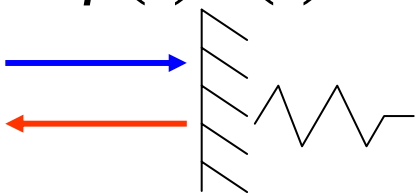
What is the energy inside?, energy balance.



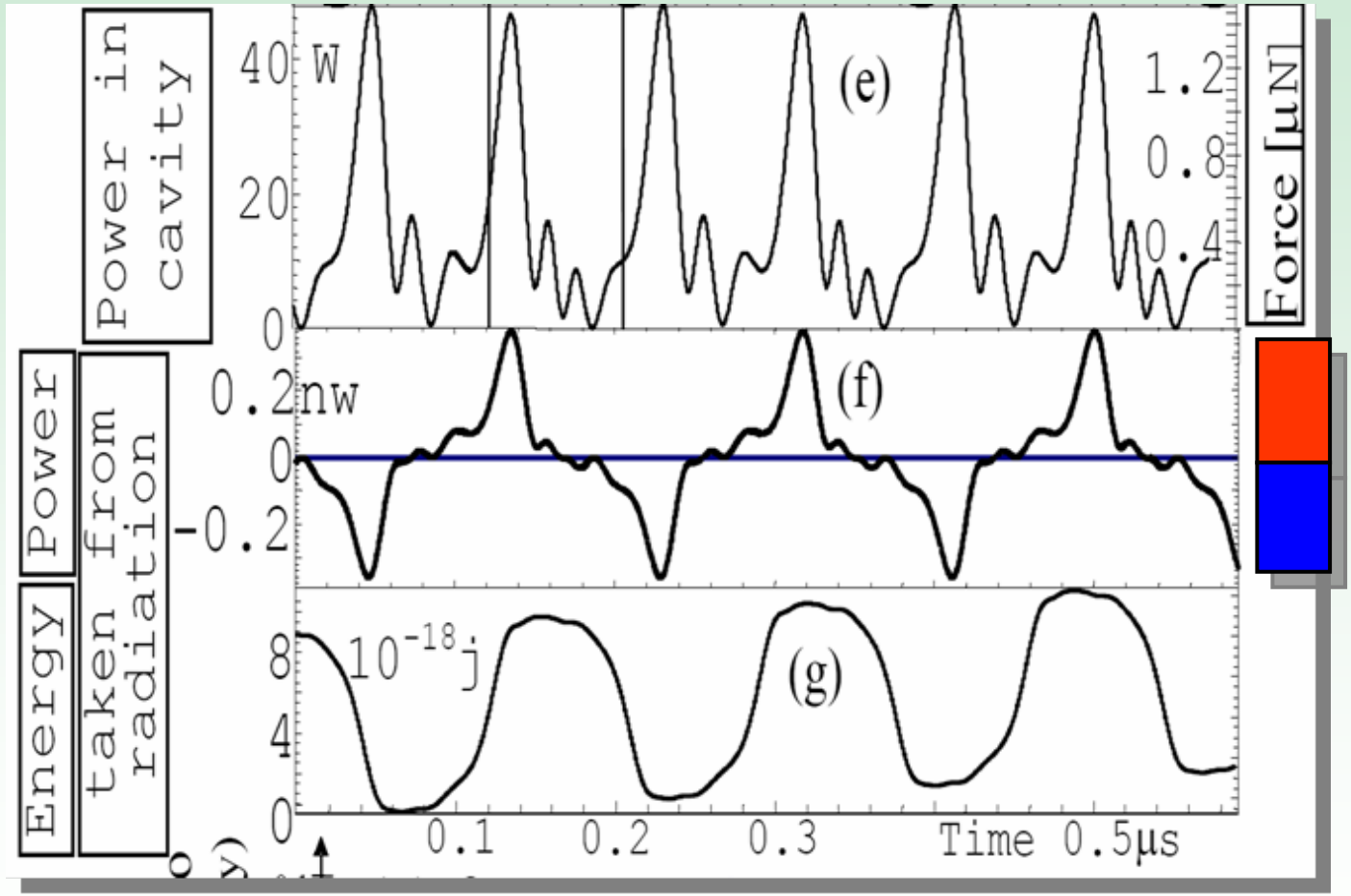


Energy dynamics

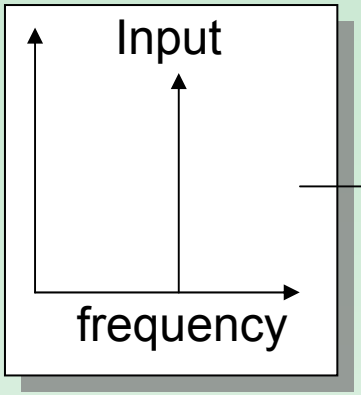
$$f(t)V(t)$$



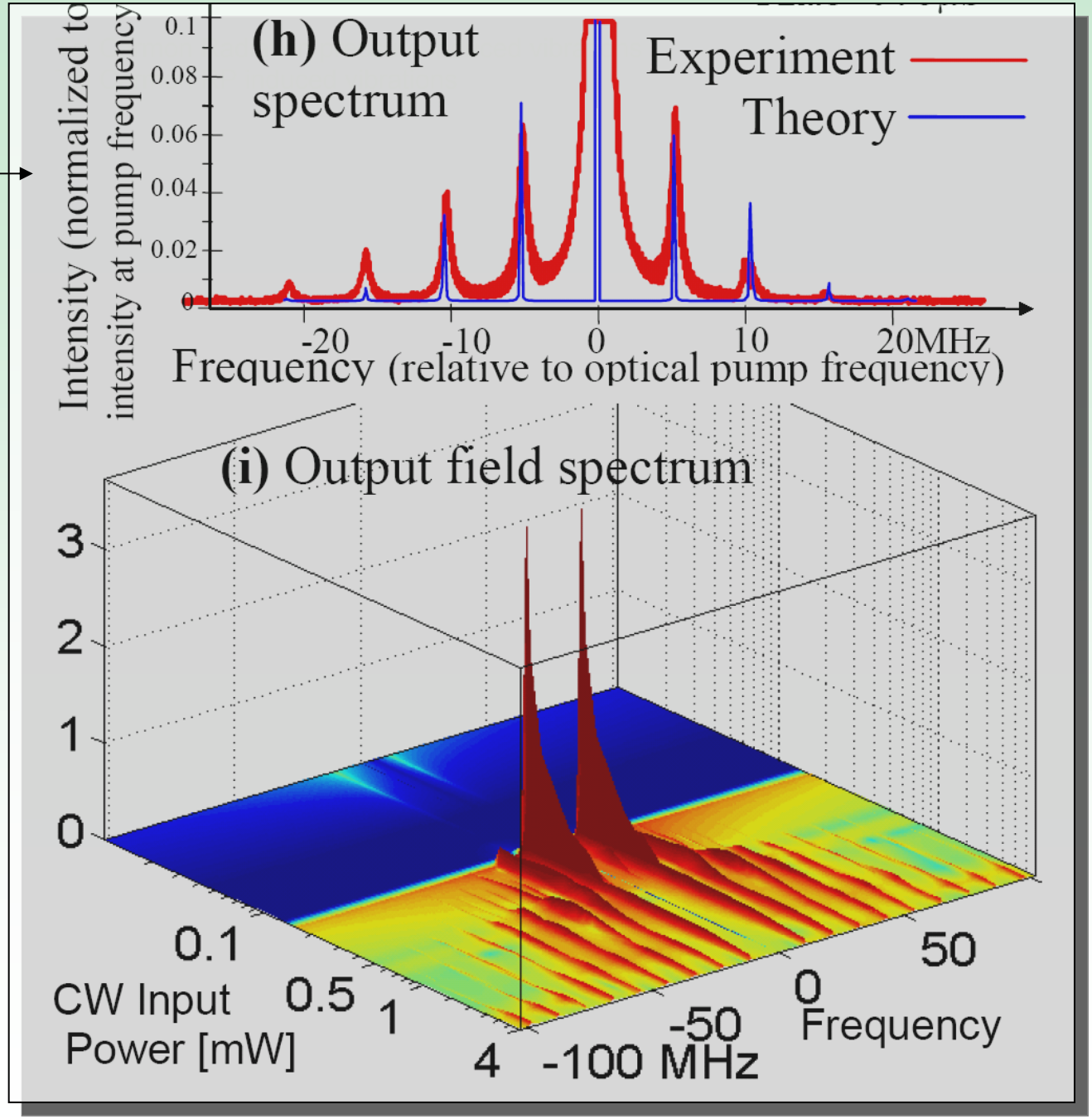
$$\int_0^t f(t')V(t')dt'$$



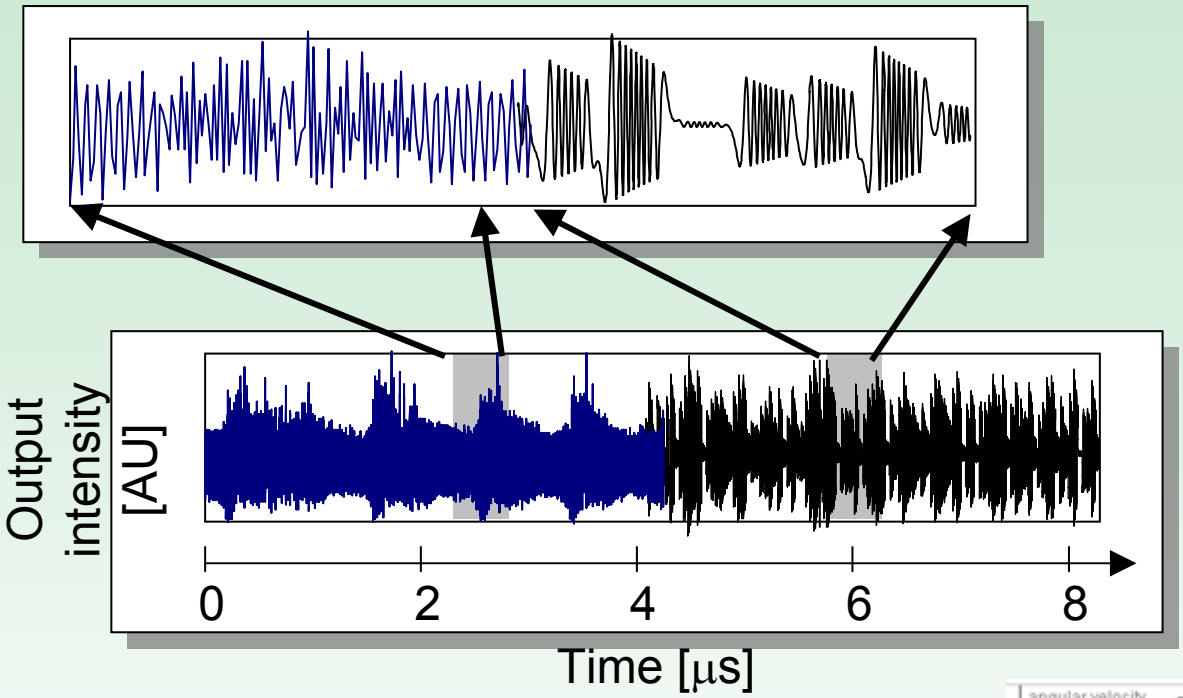
Output spectrum?



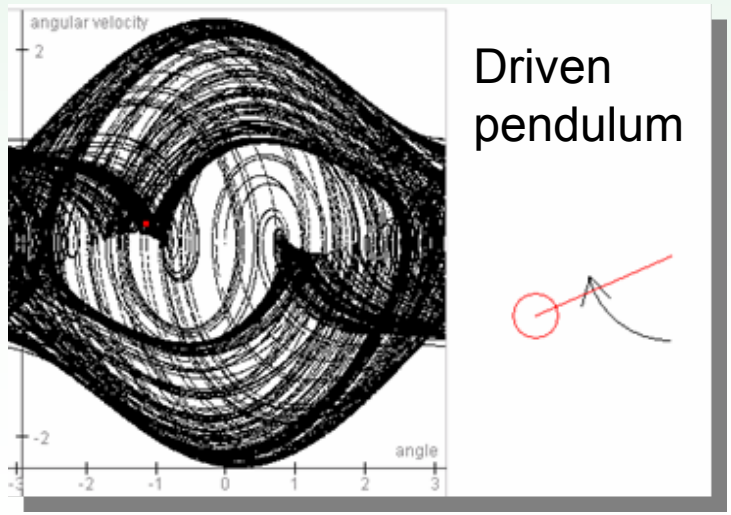
Output spectrum



Resemble chaos

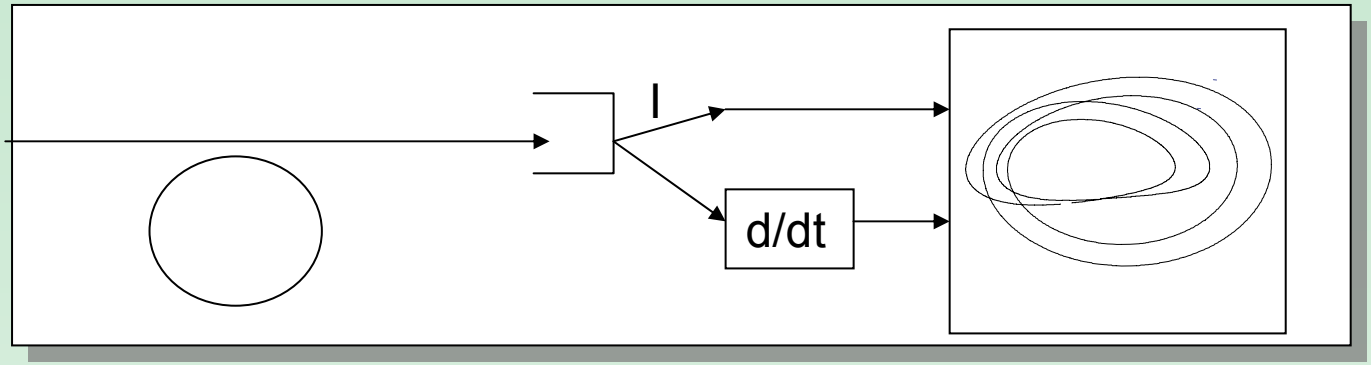


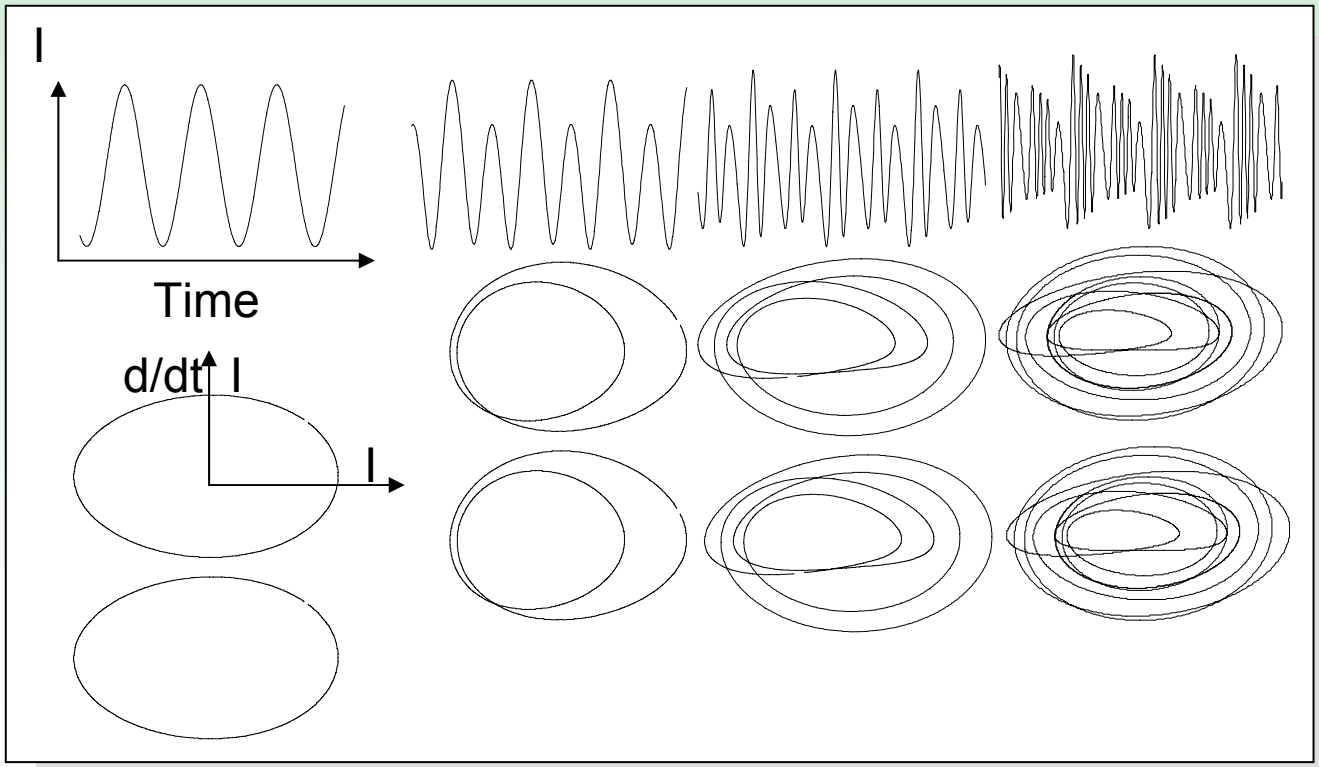
Deterministic Chaos: a system is chaotic if its trajectory through state space is sensitively dependent on the initial conditions, that is, if unobservably small causes can produce large effects.

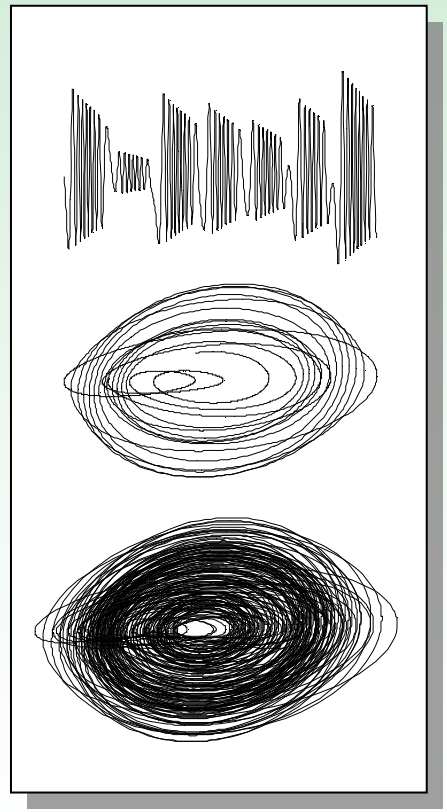


Under investigation

Phase space



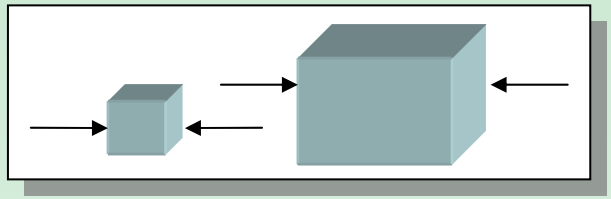




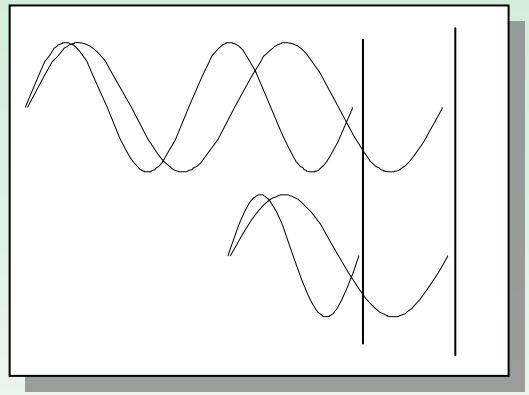
Why don't we see radiation pressure induced vibrations in large resonators?

Down scaling = sensitive to radiation pressure

1. Smaller objects flex more for the same force. (Spring constant, k=E Area/Length∝Length,)



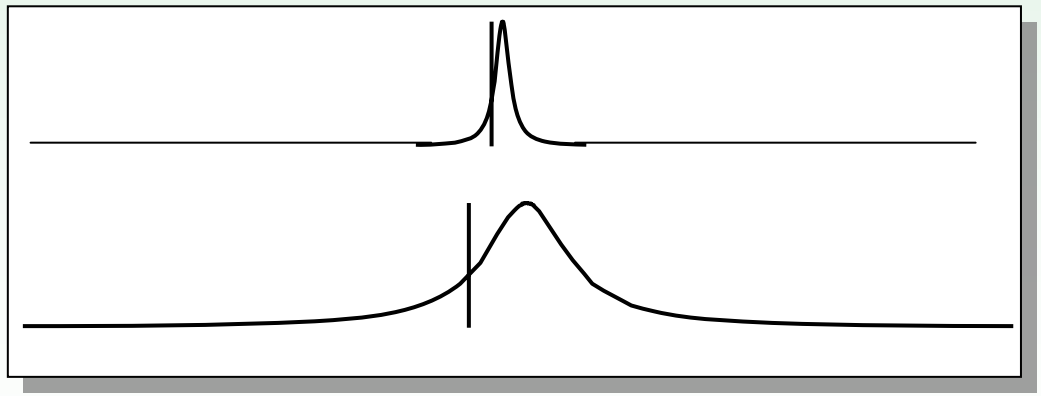
2. Smaller cavity resonance will detune more for the same flex. (change in size is absorbed by a smaller number of waves)



3. Low optical loss → Narrow bandwidth

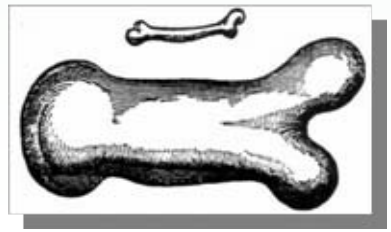
Same drift will create larger optical effect

Radiation pressure are expected in small resonators with low losses

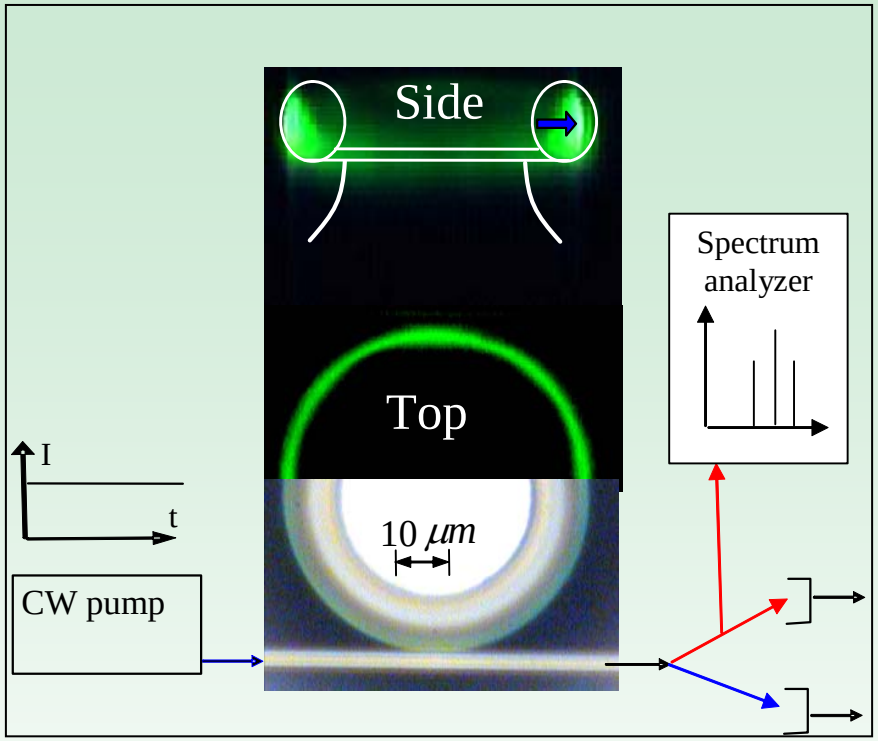


Smaller objects in nature are not just scaled replicas of similar big objects. (Galileo, Dialogue Concerning Two New Sciences, 1638)

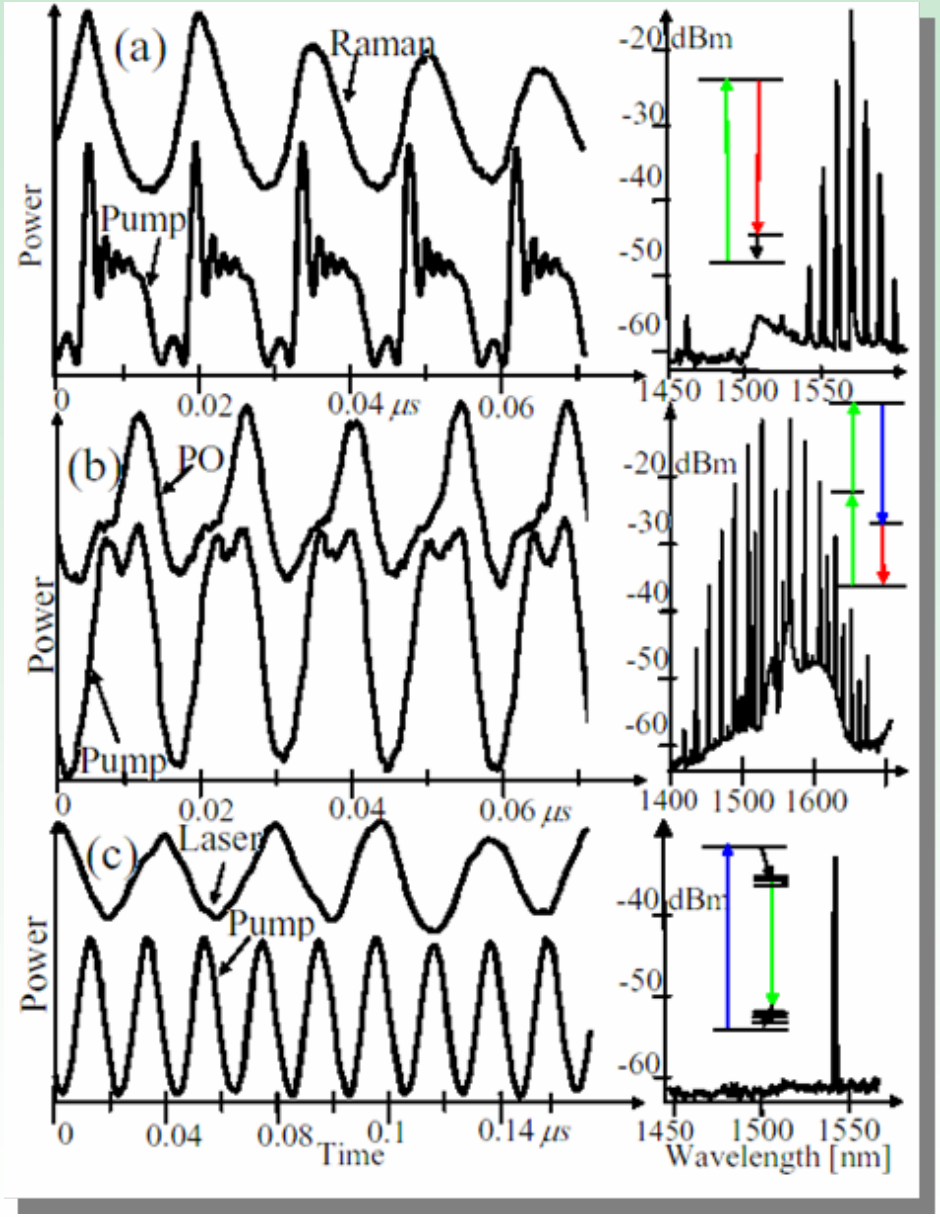
Experimental demonstration, when cavity is small enough radiation pressure effects are the first to appear



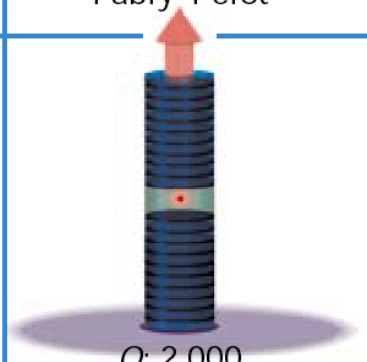
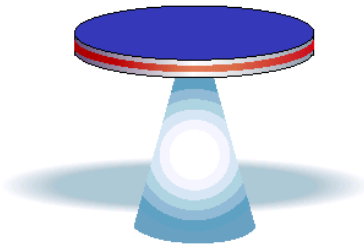
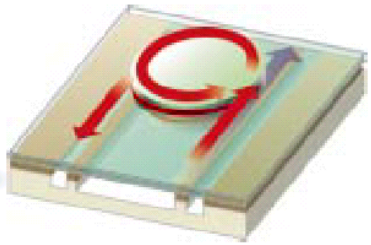
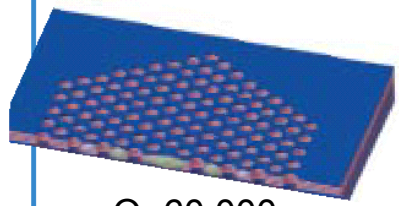
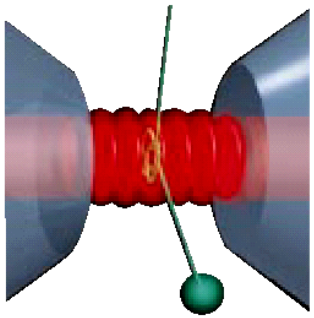
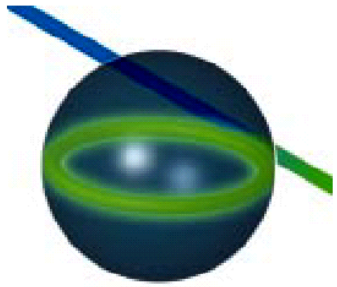
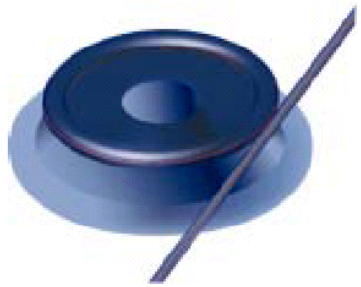
Experimental demonstration



When cavity is small enough, radiation pressure oscillations appear before other non linear effects (Raman, Kerr OPO, Er lasing)

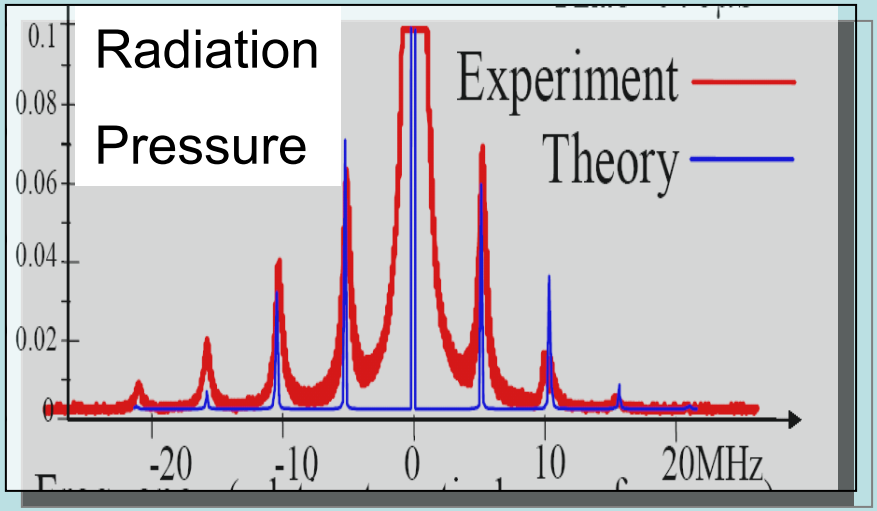


Compare with other systems

	Fabry–Perot	Whispering gallery		Photonic crystal
High Q	 Q: 2,000 V: 5 $(\lambda/n)^3$	 Q: 12,000 V: 6 $(\lambda/n)^3$	 Q_{III-V}: 7,000 Q_{Poly}: 1.3x10^5	 Q=60,000 V: 1.2 $(\lambda/n)^3$
Ultrahigh Q	 F: 4.8×10^5 V: 1,690 μm^3	 Q: 8×10^9 V: 3,000 μm^3	 Q: 10^8	

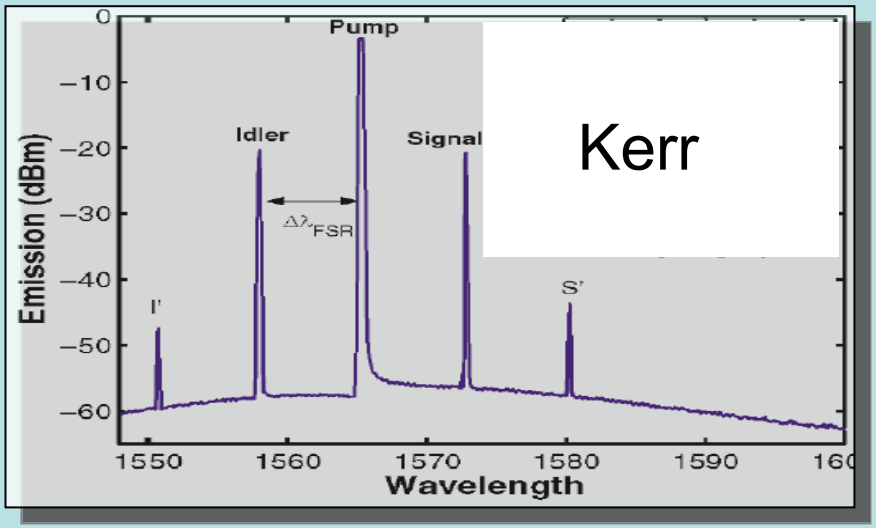
Trend= make it smaller, make it with less optical losses.

Compare with other phenomena



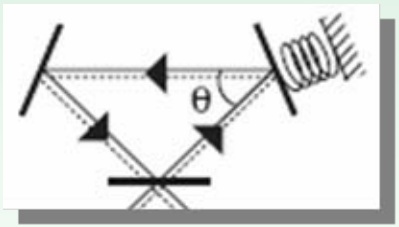
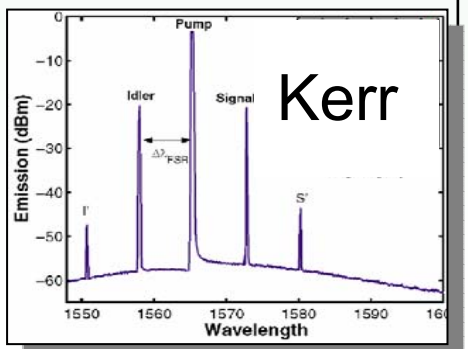
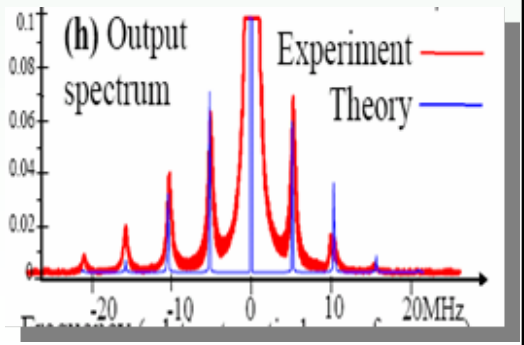
Radiation pressure parametric oscillations

10 MHz idler to signal distance



Kerr parametric oscillations
THz idler to signal distance

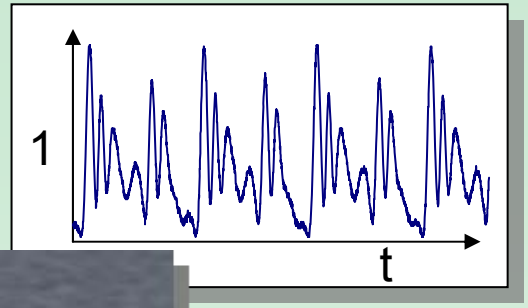
T. J. Kippenberg, S. M. Spillane,and K. J. Vahala, Phys. Rev. Lett. **93**, 083904 (2004.)

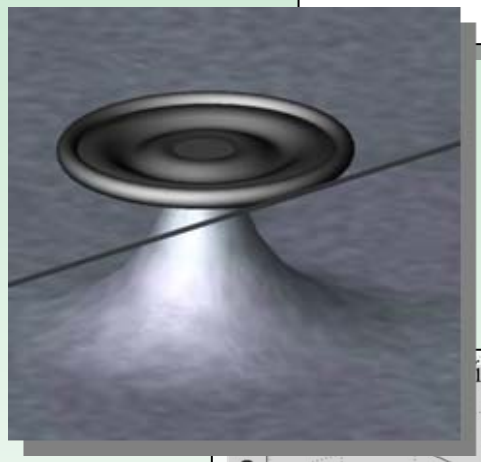
	Kerr parametric oscillations	Radiation pressure parametric oscillations
Optical path length, nL	n(I) Increases with intensity	L(I) Increases with intensity
Squeezing	L. Hilico et. al. "Squeezing with χ^3 materials" Appl. Phys. B. 55, 202 (1992)	
Entangled pair of photons	Laurat et. al., "*Characterization and optimization of entangled states produced by a self phase-locked OPO*"	•Giovannetti 
Idler to signal distance is	Optical cavity F.S.R. THz 	Mechanical eigenfrequency 10MHZ 

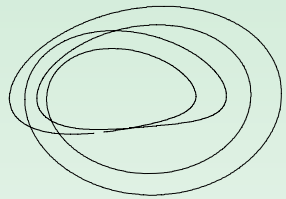


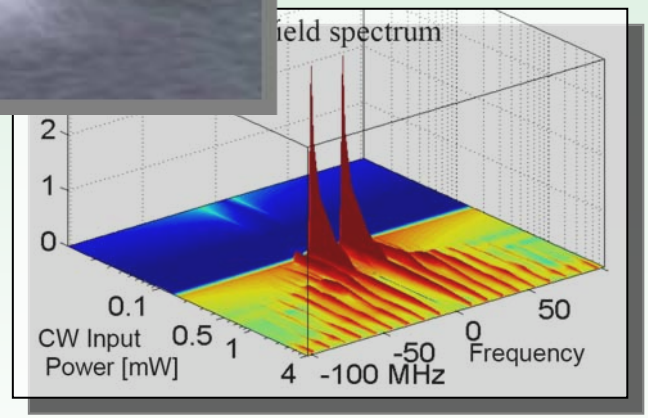
Conclusions

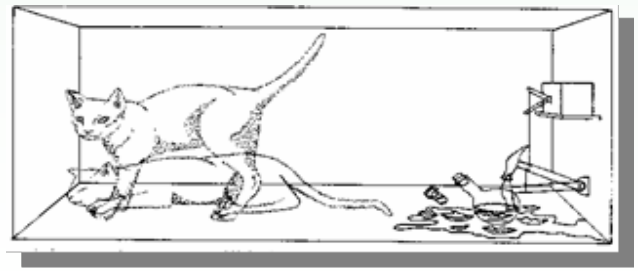
- Coupling between optical and mechanical resonances despite of the 8 orders of magnitude frequency difference.
- Oscillation is regenerative, exhibiting classic threshold behavior and requiring no external temporal modulation of the pump wave.
- When cavity is small enough, radiation pressure threshold is lower than Other nonlinearities (Raman, Kerr parametric oscillations and erbium lasing)
- No photons are lost when pushing the mirror is attractive for quantum optics (Q0~10^8, Qa~10^3, fiber coupled).











Who is interested in radiation pressure?

Appl. Phys. B 55, 202–209 (1992)

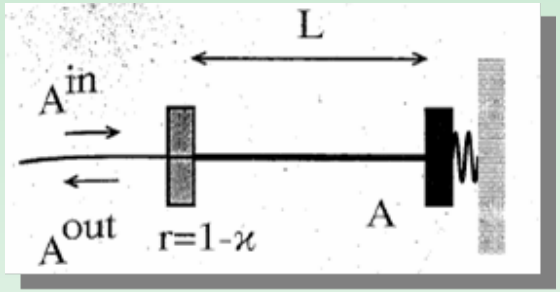
Squeezing with $\chi^{(3)}$ Materials

L. Hilico[1], J.M. Courty[1], C. Fabre[1], E. Giacobino[1], I. Abram[2], and J.L. Oudar[2]

Applied Physics B Photophysics and Laser Chemistry

© Springer-Verlag 1992

- Radiation pressure ~ Kerr nonlinearity.
- Path length increases with intensity
- Both enable squeezing



EUROPHYSICS LETTERS 1 June 2001

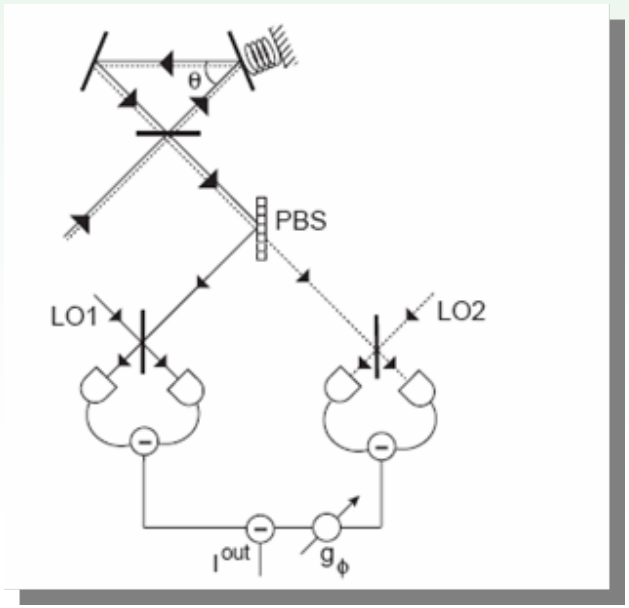
Europhys. Lett., 54 (5), pp. 559–565 (2001)

Radiation pressure induced Einstein-Podolsky-Rosen paradox

V. Giovannetti[1](*), S. Mancini[2] and P. Tombesi[1]

- Entanglement

- Radiation pressure as an ambassador of quantum properties from photons to the macroscopic world





VOLUME 88, NUMBER 12 PHYSICAL REVIEW LETTERS 25 MARCH 2002

Entangling Macroscopic Oscillators Exploiting Radiation Pressure

Stefano Mancini,[1,3] Vittorio Giovannetti,[2] David Vitali,[3] and Paolo Tombesi[3]

VOLUME 90, NUMBER 13 PHYSICAL REVIEW LETTERS week ending 4 APRIL 2003

Scheme for Teleportation of Quantum States onto a Mechanical Resonator

Stefano Mancini, David Vitali, and Paolo Tombesi

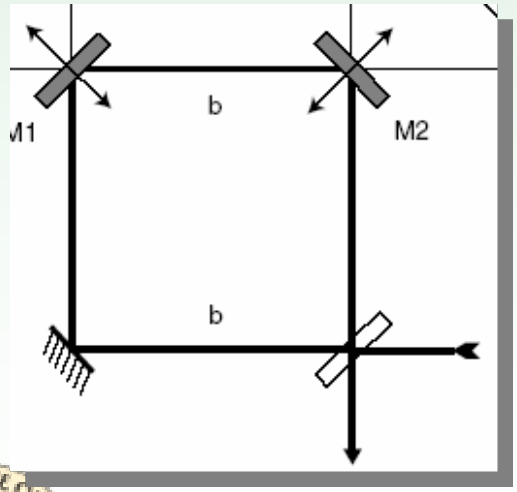
VOLUME 91, NUMBER 13 PHYSICAL REVIEW LETTERS week ending 26 SEPTEMBER 2003

Towards Quantum Superpositions of a Mirror

William Marshall,[1,2] Christoph Simon,[1] Roger Penrose,[3,4] and Dik Bouwmeester[1,2]

Macroscopic objects:

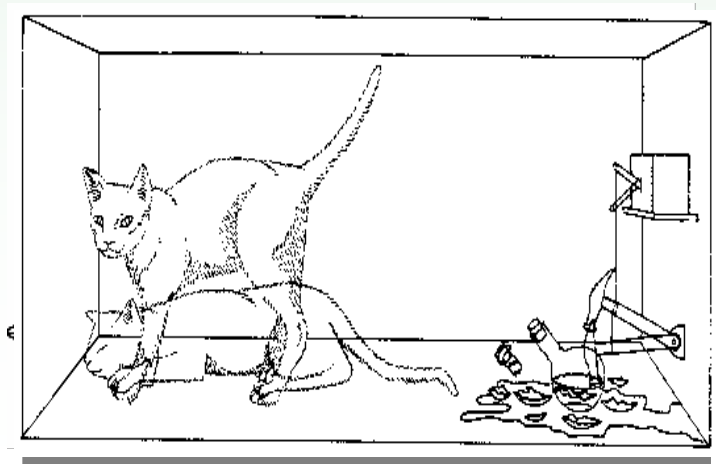
Entangled



Teletransported



Superposition of



The end.

Questions: Thermal or Radiation pressure?

Question is more than 100 years old-->



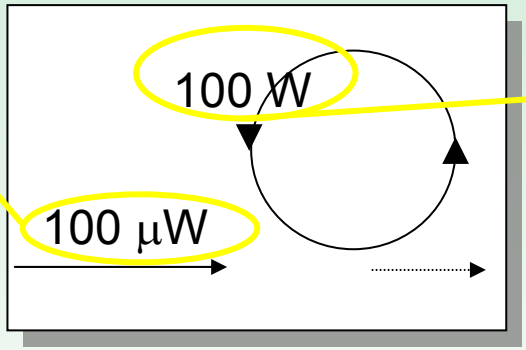
Radiation pressure induced vibrations

Thermal

$$C_p \Delta\dot{T} - K \Delta T = I$$

$$\Delta\lambda = a \Delta T$$

Thresholds α Losses



$$m \Delta\ddot{r} + b \Delta\dot{r} + k \Delta r = I$$

$$\Delta\lambda = a \Delta r$$

Thresholds α Losses^3

Thermal effect are proportional to optical loss while radiation pressure are proportional to circulating radiation;

In our cavity, radiation pressure threshold is proportional to loss^3

(Thermal threshold is expected to be proportional to loss^1)



More differences between thermal fluctuations and Radiation pressure induced vibrations.

Thermal

$$C_p \Delta\dot{T} - K \Delta T = I$$

$$\Delta\lambda = a \Delta T$$

Radiation pressure induced vibrations

$$m \Delta\ddot{r} + b \Delta\dot{r} + k \Delta r = I$$

$$\Delta\lambda = a \Delta r$$

•Two close (few FWHM) resonances are needed.

•Oscillation frequency $\propto$1/(Thermal time constant). Which is few orders of magnitude smaller than RP induced vibrations

•No distinct peak in Fourier space

•One resonance is enough.

•Oscillation frequency = Mechanical eigenmode/ .

•Distinct peak in Fourier space

•Evolution from sine to a train of decaying peaks and finally to random oscillations.

For more on thermal effects in cavities see:

•V. S. Ilchenko and M. L. Gorodetsky, "Thermal nonlinear effects in optical whispering gallery microresonators," Laser Phys. **2**, 1004-1009 (1992).

•T. Carmon, L. Yang, and K. J. Vahala, "Dynamical thermal behavior and thermal self-stability of microcavities," Opt Express **12** (20), 4742-4750 (2004).